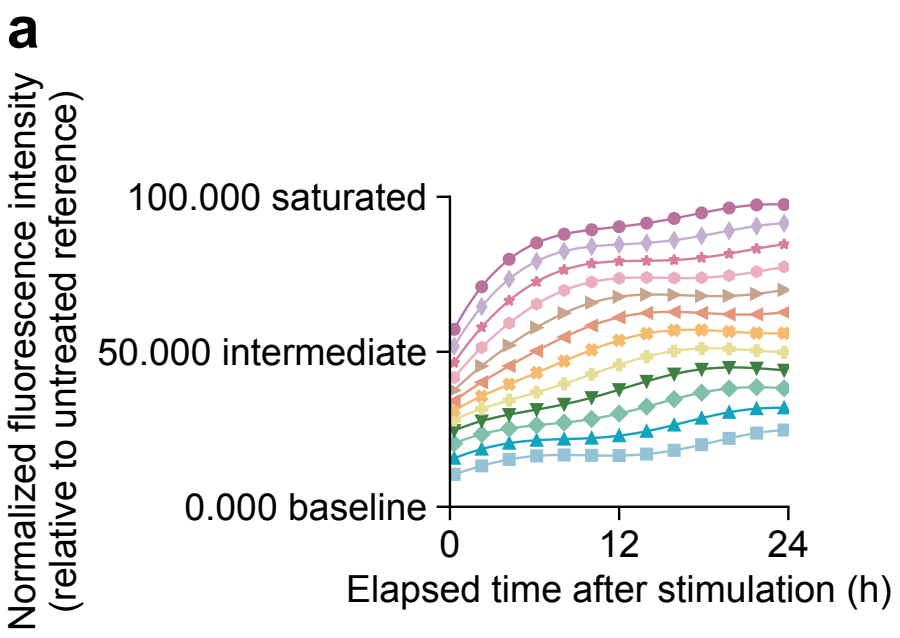
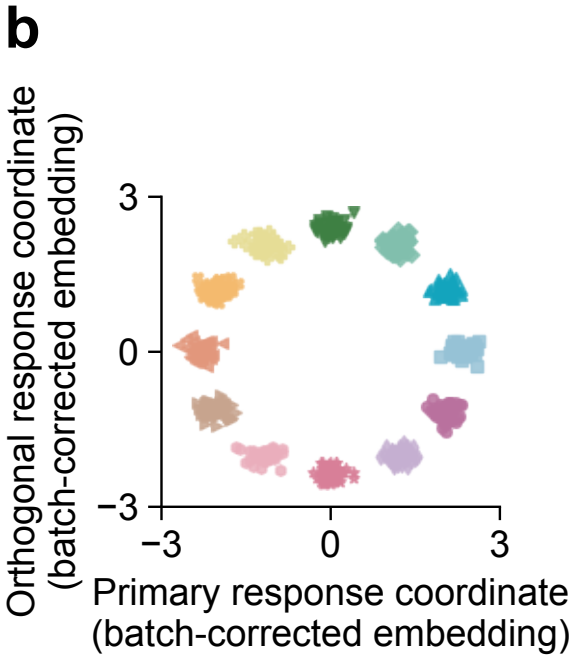
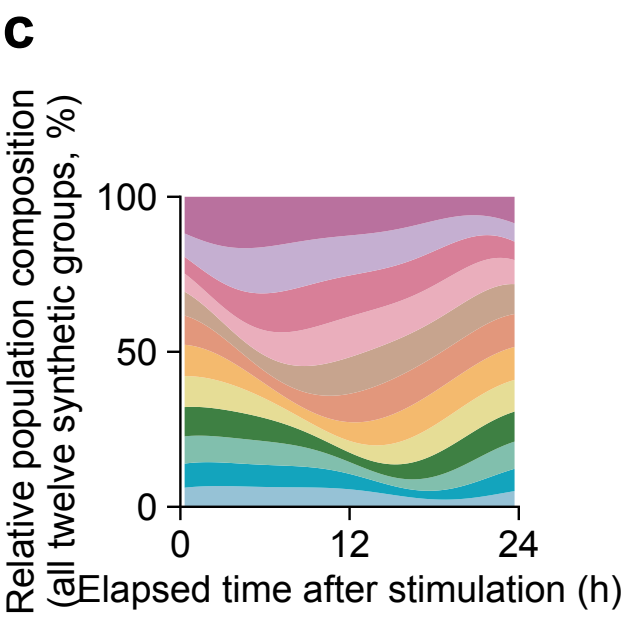
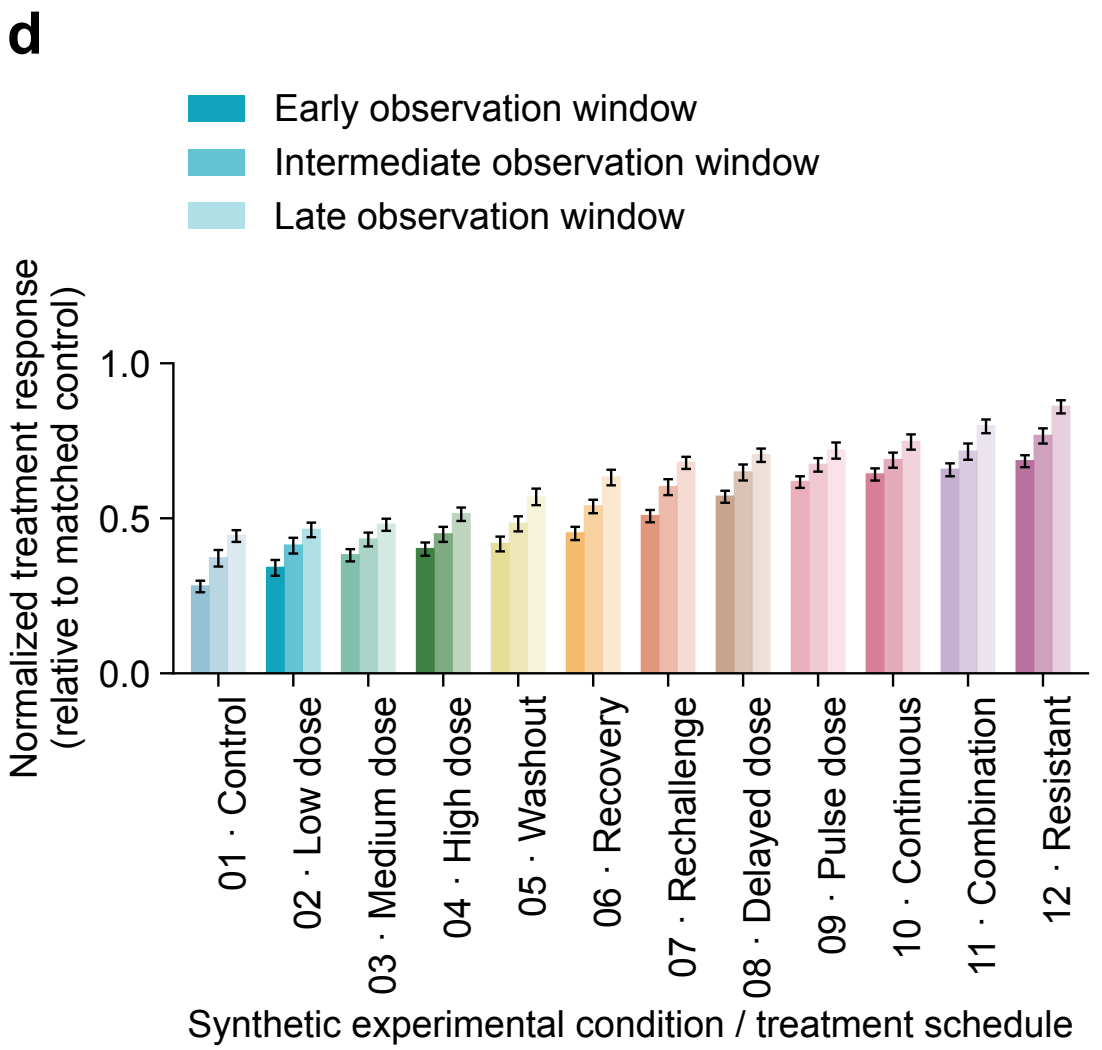
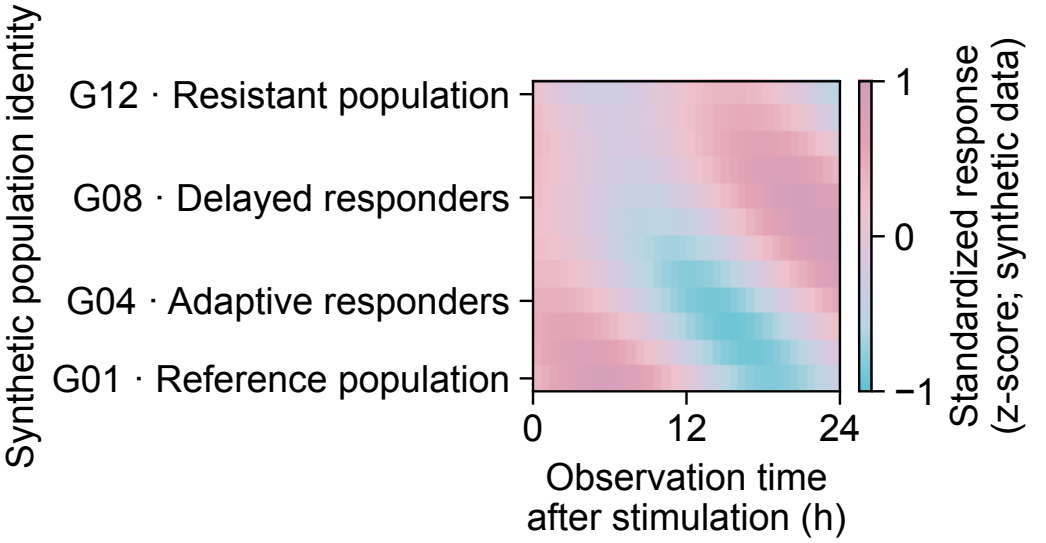
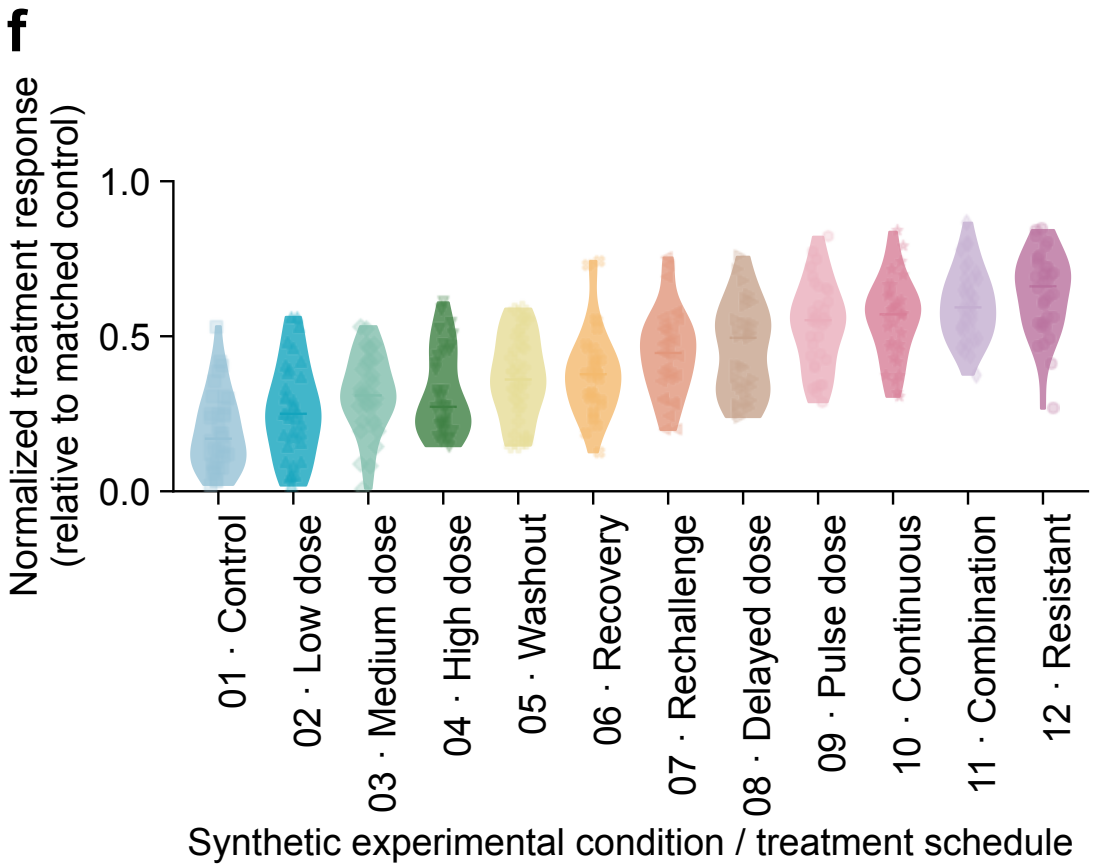

a
Normalized fluorescence intensity
(relative to untreated reference)
100.000 saturated
50.000 intermediate
0.000 baseline
0
12
24
Elapsed time after stimulation (h)

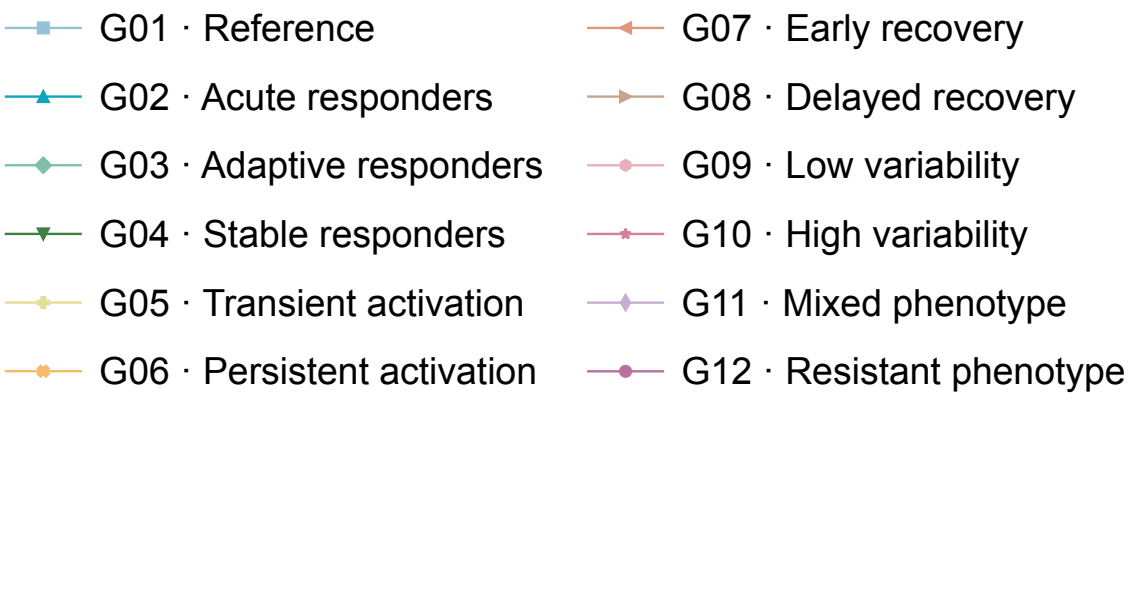

b
Orthogonal response coordinate
(batch-corrected embedding)
3
0
−3
−3
0
3
Primary response coordinate
(batch-corrected embedding)


c
Relative population composition
(all twelve synthetic groups, %)
100
50
0
0
12
24
Elapsed time after stimulation (h)


d
Early observation window
Intermediate observation window
Late observation window
Normalized treatment response
(relative to matched control)
1.0
0.5
0.0
01 · Control
02 · Low dose
03 · Medium dose
04 · High dose
05 · Washout
06 · Recovery
07 · Rechallenge
08 · Delayed dose
09 · Pulse dose
10 · Continuous
11 · Combination
12 · Resistant
Synthetic experimental condition / treatment schedule


e
Synthetic population identity
G12 · Resistant population
G08 · Delayed responders
G04 · Adaptive responders
G01 · Reference population
0
12
24
Observation time
after stimulation (h)
1
0
−1
Standardized response
(z-score; synthetic data)


f
Normalized treatment response
(relative to matched control)
1.0
0.5
0.0
01 · Control
02 · Low dose
03 · Medium dose
04 · High dose
05 · Washout
06 · Recovery
07 · Rechallenge
08 · Delayed dose
09 · Pulse dose
10 · Continuous
11 · Combination
12 · Resistant
Synthetic experimental condition / treatment schedule


G01 · Reference
G02 · Acute responders
G03 · Adaptive responders
G04 · Stable responders
G05 · Transient activation
G06 · Persistent activation
G07 · Early recovery
G08 · Delayed recovery
G09 · Low variability
G10 · High variability
G11 · Mixed phenotype
G12 · Resistant phenotype